https://xtremepape.rs/

Question	Answer	Marks
1	**Defining the problem**	
	A is the independent variable and *k* is the dependent variable **or** vary *A* and measure *k*	**1**
	keep *N* constant	**1**
	Methods of data collection	
	labelled diagram of workable experiment including: • spring fixed at one end to a support • load attached to the other end of the spring • labelled load	**1**
	method to measure mass or weight of load: use top-pan balance to measure mass or newton meter to measure weight	**1**
	use of a micrometer/calipers to determine *t* **and** rule/calipers to measure the diameter of the spring	**1**
	method to measure extension, e.g. labelled ruler drawn parallel to spring, equilibrium position and displaced position indicated and *x* indicated **or** description of use of ruler to measure equilibrium position and displaced position and difference determined	**1**
	Method of analysis	
	plot a graph of *k* against $1/A^{3/2}$ (or $A^{-3/2}$) or equivalent e.g. lg *k* against lg *A*	**1**
	relationship valid if a straight line passing through the origin is produced (for lg *k* against lg *A*, relationship valid if a straight line with gradient –3/2)	**1**
	$\beta = \dfrac{\text{gradient} \times N}{\rho t^4}$ [for lg *k* against lg *A*, $\beta = 10^{y\text{-intercept}} \times N / (\rho t^4)$]	**1**

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Question	Answer	Marks
1	**Additional detail including safety considerations**	**6**
	D1 use safety goggles/safety screen <u>to prevent injury to eyes from (moving) spring/load</u> **or** use cushion/sand box <u>in case load falls</u>	
	D2 keep t constant	
	D3 $k = \frac{mg}{x}$ or $\frac{F}{x}$	
	D4 use of set square when taking measurements to determine extension of spring	
	D5 repeat measurement of t <u>along wire/spring</u> and average	
	D6 repeat measurement of diameter D of spring (to determine A) <u>in different directions</u> and average	
	D7 use of $A = \frac{\pi D^2}{4}$	
	D8 method to ensure clamped rule to measure extension is vertical, e.g. correctly positioned set square indicated at right angles between the rule and the horizontal surface **or** plumb line shown in appropriate position	
	D9 method to determine the density of the wire **or** additional detail on construction of coil	
	D10 method to determine the mean diameter of the spring, e.g. subtract t from external diameter of spring	

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